

Skin and Eye

Patterns of care and outcomes of proton and eye plaque brachytherapy for uveal melanoma: Review of the National Cancer Database

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ABSTRACT

PURPOSE: To examine national practice patterns and outcomes of eye plaque brachytherapy compared to proton external beam radiotherapy in the treatment of choroid melanoma.**METHODS AND MATERIALS:** Demographic and clinical data for 1224 patients with choroid melanoma treated with either brachytherapy or proton beam therapy from 2004 to 2013 were obtained from the National Cancer Database. Logistic regression and propensity score matching was used to create a 1:1 matched cohort. Kaplan-Meier and Cox regression analyses were performed to evaluate survival in brachytherapy and proton groups.**RESULTS:** Median followup was 37 and 29 months for brachytherapy and protons, respectively. Most patients were treated with brachytherapy ($n = 996$) vs. protons ($n = 228$). Proton patients came from more urban, affluent, and educated zip codes, and they were more likely to be treated at an academic center (all $p < 0.004$). In the propensity-score matched cohort, 2-year overall survival was 97% vs. 93%, and 5-year overall survival was 77% vs. 51% for brachytherapy and protons, respectively ($p = 0.008$). Multivariate Cox regression found older age (hazard ratio [HR] = 1.06, 95% confidence interval [CI] = 1.03–1.09), larger tumor diameter (12–18 mm, HR = 2.48, 95% CI = 1.40–4.42, >18 mm, HR = 6.41, 95% CI = 1.45–28.35), and protons (HR = 1.89, 95% CI = 1.06–3.37) were negative prognosticators of survival.**CONCLUSIONS:** Patients selected for proton treatment have inferior survival outcomes compared to brachytherapy in this retrospective analysis. There may be unaccounted variables that influence survival, warranting further prospective studies. © 2017 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords:

Uveal melanoma; Choroid melanoma; Plaque brachytherapy; Proton therapy; National Cancer Database

Introduction

Uveal melanoma accounts for 2.9% of all melanoma cases in the United States, with an age-adjusted incidence of 4.3 per million (1). It is the most common primary intraocular tumor and predominantly affects fair-skinned individuals in their 60–70s (2, 3). The Collaborative Ocular Melanoma Study (COMS) established in 2001 that overall survival (OS) was no different between patients receiving enucleation or I-125 plaque brachytherapy, with 5- and 12-year OS of 81% and 57%, respectively (4, 5).

Plaque brachytherapy and eye-preserving therapy are now the preferred treatments of uveal melanoma in the United States (3). The American Brachytherapy Society currently recommends plaque brachytherapy for tumors that are 2.5–10 mm in depth and <16 mm in diameter. Larger tumors, gross extrascleral extension, ring melanoma, or >50% involvement of the ciliary body are contraindications to plaque brachytherapy (6). Therefore, other strategies of eye-preserving therapy have been tested. A recent Surveillance, Epidemiology, and End Results Program analysis showed no difference in 5-year OS (~83%)

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in patients treated with plaque brachytherapy compared to external beam radiotherapy (7). However, the Surveillance, Epidemiology, and End Results Program database does not distinguish between types of external beam radiotherapy or include important clinical factors such as tumor depth and size, as well as socioeconomic factors.

In the only randomized trial comparing brachytherapy to helium ion external beam radiotherapy, local control and salvage enucleation rates were better with helium ions, with no difference in OS (8, 9). However, the prescribed dose was 70 cobalt gray equivalent (CGE) to the tumor apex, which is lower than the current standard of 85 Gy for plaque brachytherapy. Proton therapy is the more popular charged-ion therapy used around the world. A number of retrospective reports of proton therapy in uveal melanoma have shown 5-year local control rates of 85–96% (10–15), but there has not been a direct comparison of proton therapy to brachytherapy. This study analyzes the national patterns of care for plaque brachytherapy and proton therapy for choroid melanoma, reporting on significant factors impacting survival.

Methods and materials

Data source and study population

The National Cancer Database (NCDB) Participant User File (PUF) is a deidentified national registry of cancer patient clinical and demographic data with survival outcomes run by the American College of Surgeons and the American Cancer Society. Approximately 70% of patients treated at Commission on Cancer-accredited cancer centers are included in the registry with standardized coding and data item definitions. NCDB is not population-based and underrepresents rural treatment centers areas and minority populations. A total of 7880 patients aged 18–90 years were identified who presented with choroid melanoma (code 693 from *International Classification of Diseases for Oncology*, third edition) from 2004 to 2013 as their only malignancy. Patients with nodal or metastatic disease, incomplete staging information including basal diameter and tumor thickness, or received surgery or chemotherapy were excluded. Radiation therapy was specified as either brachytherapy or proton therapy for a final total cohort of 1224 patients (Fig. 1).

Demographic and clinical data included age, sex, treatment facility type, treatment year, race, insurance status, Charlson-Deyo comorbidities score, distance to treatment, time to starting radiation, and socioeconomic status of the patient's home address were included. The American Joint Committee on Cancer (AJCC), seventh edition, was introduced in 2010 and included differences in T staging. Due to this shift in staging in the study period, all patients were restaged according to the American Joint Committee on Cancer, seventh edition, from the reported basal diameter and tumor thickness. Ciliary or extraocular extension and proton dose are also reported. The PUF only has

deidentified data and was exempt from institutional review board oversight.

Statistical analysis

The χ^2 and Fisher exact tests were used to compare categorized demographic and clinical variables in brachytherapy and proton groups. Student's two-tailed *t* test was used to compare differences in age and average tumor dimensions. Overall survival was calculated from diagnosis until death with censoring of patients who are still alive at last followup. Significance of Kaplan-Meier estimates of OS was determined by log-rank test. Cox proportional hazards analysis was used for multivariate analysis of predictors of OS. All variables were entered in a forward-conditional manner into the model and significance was set as *p*-value < 0.05.

Propensity scores and a 1:1 matched cohort of brachytherapy and proton patients were created as previously described (16). Differences in demographic and clinical variables in the brachytherapy and proton groups were identified by logistic regression and distilled into a propensity score for receiving proton therapy. Patients with matching propensity scores were identified in the larger brachytherapy control group for each proton patient. This smaller cohort of 452 patients (226 in each group) was separately analyzed with Kaplan-Meier and Cox proportional hazards analyses as above. Propensity score matching was performed with the MatchIt function (17) in R, version 3.2.3 (2015 The R Foundation for Statistical Computing). All other analyses were done in SPSS Statistics, version 23 (IBM Corporation, Armonk, NY).

Results

Patient demographics

A total of 1224 patients were included in the initial analysis. Median followup was 37 and 29 months for brachytherapy (*n* = 996) and protons (*n* = 228), respectively. Many records of tumor dimensions were incomplete before 2010 and not included. Thus, 93% of patients in this cohort were treated from 2010 to 2013. The mean age was 61 years (range 20–90 years), and 52% were male. Most patients were white (95%), insured (98%), and had no other major comorbidities (83%). The majority of patients traveled more than 60 miles to get treatment (74%) and most were treated within 60 days of diagnosis (87%). Median proton dose was 56 CGE in four fractions (range 50–70.4 in four–seven fractions). Brachytherapy prescription doses were not specified. There were no significant differences in age, sex, insurance status, Charlson-Deyo score, T-stage, ciliary/extraocular extension, or tumor thickness. Patients receiving protons were more likely to be treated at an academic center (99% vs. 76%) with more experience (>50 patients 90% vs. 56%), nonwhite (86% vs. 96%), traveled >60 miles to treatment (81% vs. 72%), started treatment >60 days from diagnosis (19% vs. 11%), and had tumors with basal diameter ≤12 mm (69%

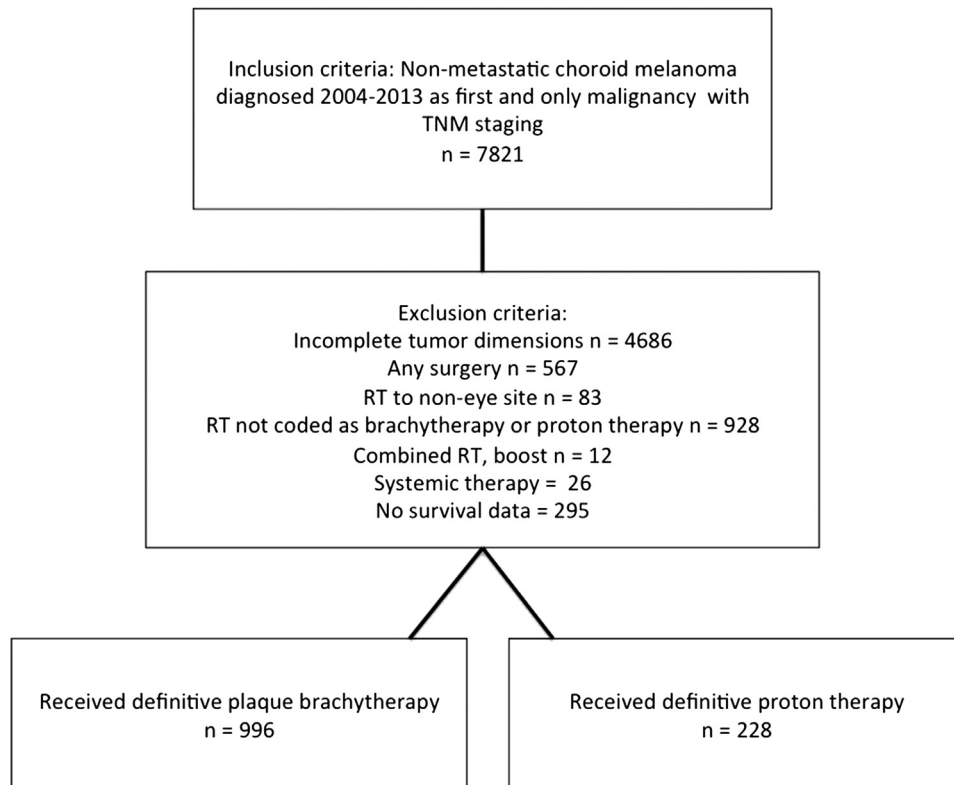


Fig. 1. Consort diagram. RT = radiation therapy.

vs. 60%); proton patients also came from more urban, affluent, and educated zip codes (all $p < 0.05$). Patient demographic and tumor characteristics are summarized in Table 1.

Logistic regression and propensity score matching

Multivariate logistic regression was done to identify variables that were associated with receipt of protons vs. brachytherapy. All clinical and demographic factors listed in Table 1 were included in the analysis. Significant variables predicting proton treatment included treatment at an academic center (odds ratio [OR] = 39.8), nonwhite race (OR = 6.5), traveling more than 60 miles to treatment (OR = 2.5), and starting treatment more than 60 days from diagnosis (OR = 2.5), all p -values < 0.001 . Negative predictors for proton treatment included having a home address where median income $< \$38,000$ (OR = 0.26), high school education rate $< 21\%$ (OR = 0.32), and population $< 250,000$ (OR = 0.38). Propensity scores were calculated for the above significant variables in SPSS. Patients treated with protons ($n = 226$) were matched in a 1:1 fashion with those treated with brachytherapy ($n = 226$) using propensity scores. In the matched cohort, persistent differences remained with respect to race (white race, protons 86% vs. brachytherapy 93%, $p = 0.046$) and treatment center experience (> 50 patients, protons 89% vs. 67%, $p < 0.001$). These patient demographic and tumor characteristics are summarized in Table 2.

Overall survival outcomes

Kaplan-Meier estimates of 2-year OS for the whole cohort were 96% vs. 92% for brachytherapy and protons, respectively; 5-year OS was 81% vs. 54%, respectively ($p < 0.001$) (Fig. 2a). On multivariate analysis, the following variables were associated with increased risk of mortality: older age (hazard ratio [HR] = 1.05), larger tumor diameter (12–18 mm, HR = 2.25, > 18 mm, HR = 3.56), treatment at an academic facility (HR = 2.07), and protons (HR = 1.91) (all $p < 0.02$) (Table 3). In the propensity-score matched cohort, 2-year OS was 97% vs. 93%, and 5-year OS was 77% vs. 51% for brachytherapy and protons, respectively ($p = 0.008$) (Fig. 2b). On multivariate analysis, older age (HR = 1.06), larger tumor diameter (12–18 mm, HR = 2.48, > 18 mm, HR = 6.41), and protons (HR = 1.89) were associated with increased risk of mortality (all $p < 0.02$) (Table 4).

Discussion

This comparison of plaque brachytherapy and proton therapy for uveal melanoma in the NCDB found older age, larger basal diameter, and treatment with proton therapy predicted overall mortality. Baseline characteristics of patients treated with proton therapy tended to be favorable.

Table 1
Demographics and clinical characteristics

	Total (n = 1224)	Brachytherapy (n = 996)	Protons (n = 228)	p-value
Age (years)	61.0 ± 13.7	61.0 ± 13.3	60.6 ± 13.0	0.637
Sex (male)	641 (52%)	518 (52%)	123 (54%)	0.608
Academic center	916/1142 (80%)	707/931 (76%)	209/211 (99%)	<0.001
Treatment				
2004–2010	83 (7%)	81 (8%)	2 (1%)	<0.001
2010–2013	1141 (93%)	915 (92%)	226 (99%)	
White race	1157 (95%)	960 (96%)	197 (86%)	<0.001
Uninsured	31 (2%)	28 (3%)	3 (1%)	0.381
Private	662 (55%)	533 (54%)	129 (57%)	
Government	514 (43%)	419 (43%)	95 (42%)	
Treatment center experience				
>20 patients	972 (79%)	768 (77%)	204 (90%)	<0.001
>50 patients	757 (62%)	553 (56%)	204 (90%)	<0.001
Median income <\$38000	164/1222 (13%)	150/994 (15%)	14/228 (6%)	<0.001
High school education <21%	1077/1222 (88%)	890/994 (90%)	187/228 (82%)	0.003
Population <250000	447/1188 (38%)	393/970 (40%)	54/218 (25%)	<0.001
Distance to treatment >60 mi	902/1222 (74%)	717/994 (72%)	185/228 (81%)	0.006
Time to RT start >60 d	153/1224 (13%)	109/996 (11%)	44/228 (19%)	0.001
Charlson-Deyo score				
0	1017 (83%)	830 (83%)	187 (82%)	0.229
1	175 (14%)	137 (14%)	38 (17%)	
≥2	32 (3%)	29 (3%)	3 (1%)	
T1	454 (37%)	363 (36%)	91 (40%)	0.660
T2	443 (36%)	360 (36%)	83 (36%)	
T3	239 (20%)	199 (20%)	40 (18%)	
T4	88 (7%)	74 (8%)	14 (6%)	
Ciliary or extraocular extension	118/1104 (11%)	100/880 (11%)	18/224 (8%)	0.182
Basal diameter	10.9 ± 7.3	11.0 ± 7.9	10.5 ± 4.3	0.249
≤12 mm	753 (62%)	595 (60%)	158 (69%)	0.021
12.1–18 mm	429 (35%)	367 (37%)	62 (27%)	
>18 mm	42 (3%)	34 (3%)	8 (4%)	
Thickness	5.8 ± 7.6	5.8 ± 7.9	5.5 ± 6.1	0.593
≤6 mm	888 (72%)	728 (73%)	160 (70%)	0.175
6.1–9 mm	193 (16%)	151 (15%)	42 (18%)	
9.1–15 mm	99 (8%)	77 (8%)	22 (10%)	
>15 mm	44 (4%)	40 (4%)	4 (2%)	
Median proton dose (range)			56 Gy (50–70.4)	

Proton patients came from more urban, educated, and wealthy neighborhoods, and almost all were treated at an academic facility. A greater proportion of proton patients had tumors with a basal diameter ≤12 mm (69% vs. 60%, $p = 0.02$). Yet, 2-year OS was 97% vs. 93%, and 5-year OS was 77% vs. 51% for brachytherapy and protons, respectively ($p = 0.008$). Proton patients were more likely to be nonwhite, but race has not previously been found to be prognostic in uveal melanoma (18). The early mortality differences found with a median followup of 37 and 29 months for brachytherapy and protons, respectively, are surprising and may reflect undocumented comorbidities or early/occult metastases in the proton group.

The survival curve of this NCDB brachytherapy cohort mirrors that of the COMS study, which had a 5-year OS of 82% (4). Techniques for placing plaque brachytherapy have improved since the introduction of intraoperative ultrasonography. In one center, intraoperative ultrasonography changed positioning of 36% of eye plaques in order for there to be adequate margin coverage (>1 mm) (19).

Ultrasonography also allows detection of plaque tilt and supplemental treatment with transpupillary thermotherapy to boost treatment to the highest risk patients. Local tumor recurrence was 3.6% in these patients (20). However, the adoption of ultrasound techniques in the United States and then translation of excellent local control to OS is not clear. Therefore, OS in this NCDB cohort has not improved from the COMS study period of 1987–1998.

A review of prior centers' experience with proton therapy suggests that the early mortality in this NCDB cohort is abnormally high. In the largest published cohort, 3088 patients treated from 1975 to 2005 at the Massachusetts Eye and Ear Infirmary had a 15-year all-cause mortality of 49% and melanoma-related mortality of 24.6% (21). The peak rate of death was 3 years after treatment and the melanoma-specific death rate at 5 years was about 15% (21). The Nice Teaching Hospital and Curie Institut in France treated 2292 uveal melanoma patients with 60 CGE proton therapy in four fractions with a 5-year OS rate of 79% (11, 14). By comparison, we report a 5-year

Table 2
1:1 matched propensity score cohort

	Brachytherapy (n = 226)	Protons (n = 226)	p-value
Age (years)	61.0 ± 13.5	60.6 ± 13.0	NS
Sex (male)	123 (54%)	121 (54%)	NS
Academic center	204/208 (98%)	207/209 (99%)	NS
Treatment 2010–2013	226 (100%)	226 (100%)	NS
Treatment center experience			
>20 patients	183 (81%)	202 (89%)	0.017
>50 patients	151 (67%)	202 (89%)	<0.001
White race	209 (93%)	195 (86%)	0.046
Uninsured	1 (0%)	3 (1%)	NS
Private	126 (56%)	128 (57%)	
Government	99 (44%)	95 (42%)	
Median income <\$38000	12 (5%)	13 (6%)	NS
High school education <21%	201 (89%)	186 (82%)	NS
Population < 250000	55/213 (26%)	52/216 (24%)	NS
Distance to treatment >60 mi	188 (83%)	183 (81%)	NS
Time to RT start >60 days	32 (14%)	43 (19%)	NS
Charlson-Deyo score			NS
0	186 (82%)	185 (82%)	
1	34 (15%)	38 (17%)	
≥2	6 (3%)	3 (1%)	
T1	96 (42%)	89 (39%)	NS
T2	76 (34%)	83 (37%)	
T3	40 (18%)	40 (18%)	
T4	14 (6%)	14 (6%)	
Ciliary or extraocular extension	20 (9%)	18 (8%)	NS
Basal diameter	9.9 ± 4.5	10.6 ± 4.3	NS
≤12 mm	155 (69%)	156 (69%)	
12.1–18 mm	68 (30%)	62 (27%)	
>18 mm	3 (1%)	8 (4%)	
Thickness	6.1 ± 10	5.5 ± 6.1	NS
≤6 mm	167 (74%)	158 (70%)	
6.1–9 mm	34 (15%)	42 (18%)	
9.1–15 mm	16 (7%)	22 (10%)	
>15 mm	9 (4%)	4 (2%)	
Median proton dose		56 Gy (50–70.4)	

survival of 51%. Separation in the survival curves between proton and brachytherapy began 2 years after treatment, within the median followup time. Thus, longer followup time may diminish but not eliminate the separation in survival curves.

Undocumented comorbidities were possibly not balanced between the two groups. Only 1–3% of the studied patients had Charlson-Deyo scores ≥2 (or ≥4 total, since the primary cancer is not coded as a comorbidity), which seems abnormally low for this age population. The Charlson-Deyo score gives weighted scores to different comorbidities ranging from diabetes, dementia, to acute myocardial infarction,

and cancer as classified by the International Classification of Diseases, ninth revision, codes to predict 1-year mortality (22, 23). Predicted mortality ranges from 5.6% with a score of two to 23.6% for a score ≥6 (24). In the Nice Teaching Hospital series, cause of death was from melanoma (51%), other cancer (20%), or other disease (39%) (11). Therefore, comorbidities in this age group contribute significantly to OS, and more information on comorbidities and cause-specific survival will be important in future comparison studies of proton and brachytherapy.

There may have been an imbalance in patients with occult or early metastases. Despite excellent local control with the best available treatments of surgery or radiation (2), 34% of patients still develop distant metastases by 10 years after primary treatment (25), most commonly to the liver and lung (26). The median survival after metastasis is on the order of 6 months (25). Recently there has been validation of a panel of genetic markers that predict the risk of metastasis. The DecisionDx-UM gene expression profile identifies a high risk group, with 26–37% developing metastases by 3 years (27, 28). There has also been interest in detecting occult metastases at diagnosis, especially from circulating tumor cells (29). An immunomagnetic assay detected occult metastases to the bone marrow and peripheral blood in 29.9% and 1.6% of 328 patients, respectively (30). The development of molecular risk profiles for uveal melanoma will likely be an important clinical tool in the advancement of treatment.

If occult metastases are present at diagnosis, then radiation dose may play a role in stimulating the immune system with cancer neoantigens (31). The median and most commonly given dose in the NCDB proton group was 56 CGE in four fractions (range 50–70.4 CGE). Proton dose was investigated in a randomized trial comparing 50 CGE to 70 CGE for uveal melanoma at the Massachusetts Eye and Ear Infirmary. There were no differences in 5-year local recurrence (2% vs. 3%) or distant metastasis (7% vs. 8%), but dose reduction did reduce visual field loss (32). The advantage of brachytherapy is that dose can be escalated with less concern for optic nerve toxicity. Brachytherapy dose was not specified in the NCDB PUF, but the standard dose of 85 Gy to the tumor apex means there is increasing gradient of dose to the tumor base (20). Preclinical studies have shown a dose–response of the abscopal effect with increasing radiation dose above 60 biological effective dose (33), but there has not been a comparison of the homogeneous dose with proton therapy vs. heterogeneous dose of brachytherapy. Immunotherapy has been a recent breakthrough in treatment of cutaneous melanoma (34, 35), and trials in uveal melanoma may show similar benefit.

Treatment center experience is likely important, but did not play a role in decreased survival in this NCDB cohort. In an international survey of major proton treatment centers that specialize in treatment of ocular tumors, seven out of 10 treated more than 100 eye patients a year (36). Nearly all patients treated with proton therapy in the NCDB received care at academic centers in the United States,

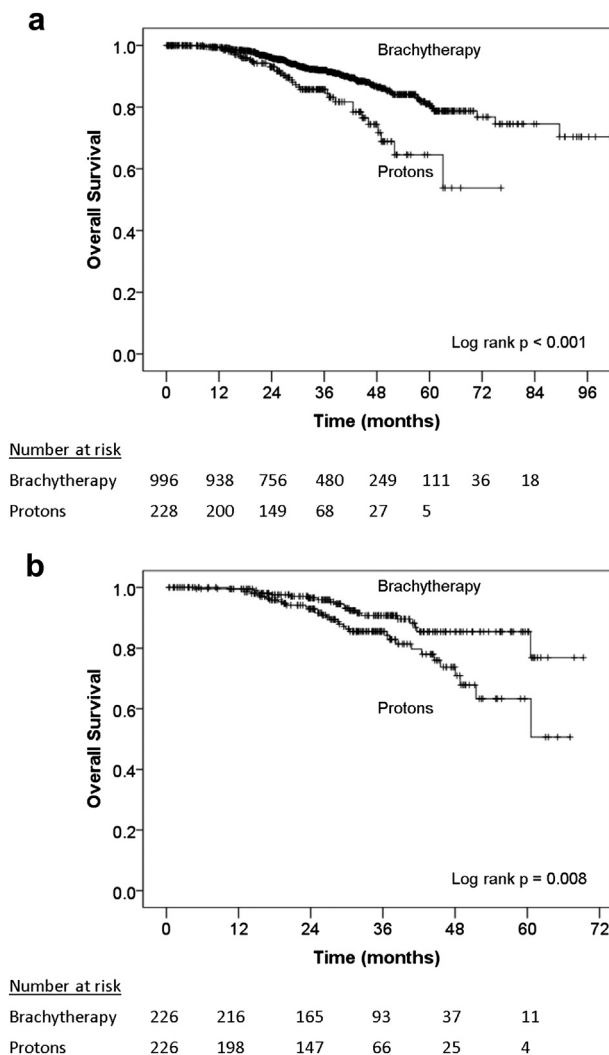


Fig. 2. Kaplan-Meier curves of overall survival for (a) whole cohort and (b) propensity matched cohort.

and 90% were treated at experienced centers (>50 uveal melanoma patients treated). Egger *et al.* treated 2435 uveal melanoma patients with 60 CGE proton therapy in four fractions and showed a 10-year cause-specific survival rate of 72.6% for patients with controlled tumors vs. 47.5% for those with recurrent tumors. Interestingly, their 5-year local control rate improved from 90.6% in the first 5 years of their experience to 98.9% in the last 5 years of their 15 years

Table 3
Cox proportional hazards multivariate model for whole cohort

	HR (95% CI)	p-value
Age	1.05 (1.04–1.07)	<0.001
Academic center	2.07 (1.15–3.71)	0.015
Basal diameter		
≤12 mm	Reference	
12.1–18 mm	2.25 (1.54–3.27)	<0.001
>18 mm	3.56 (1.25–10.1)	0.017
Proton therapy	1.91 (1.24–2.95)	0.003

HR = hazard ratio; CI = confidence interval.

Table 4
Cox proportional hazards multivariate model after propensity score matching

	HR (95% CI)	p-value
Age	1.06 (1.03–1.09)	<0.001
Basal diameter		
≤12 mm	Reference	
12.1–18 mm	2.48 (1.40–4.42)	0.002
>18 mm	6.41 (1.45–28.35)	0.014
Proton therapy	1.89 (1.06–3.37)	0.031

HR = hazard ratio; CI = confidence interval.

experience, indicating there may be a learning curve to proton therapy expertise (15). Referral patterns and patient selection may differ between this NCDB cohort and other published series. Proton patients had to travel further and start treatment later than brachytherapy patients, which may contribute to worse outcomes and poor followup.

Major limitations of this NCDB analysis include missing information on initial patient selection criteria that could impact survival. For example, T-stage does not take into account the location of the tumor, which may have prognostic value. Proton therapy may have been offered to patients who were not eligible for brachytherapy due to proximity to the optic disc. In this case, a comparison of proton and enucleation may be more fair. Initial staging scans and gene expression panels predicting risk of metastasis were also not specified, which could change the detection rate of distant metastases at diagnosis or in early followup. Lastly, melanoma-specific mortality could not be determined. When melanoma-specific mortality accounts for only 50% of overall mortality, more details on comorbidities than the Charlson-Deyo score are needed. Caution should be taken in interpreting these results due to the short median followup time and small number of proton patients compared to larger institutional reports. This report should not discourage patients from seeking opinions on proton treatment at large referral centers, especially for large tumors not eligible for brachytherapy. Patient selection and unaccounted comorbidities likely factor into the survival differences seen.

Conclusions

Brachytherapy is the more commonly used treatment for uveal melanoma in the United States. Patients selected for brachytherapy have better OS compared to proton therapy. NCDB research has many limitations, and unaccounted confounders likely explain the differences in survival outcomes between brachytherapy and proton therapy. Prospective comparative studies comparing outcomes after brachytherapy or proton treatment should be supported.

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